

Lab-6: Finite difference for PDE in Python

The aim of this lab is to apply the Finite Difference Method to find approximate solutions to Partial Differential Equations (PDE).

Instructions:

- Clone your repository from the UPPA gitlab: it contains only the class `Mesh1D` for the mesh.
- Do one .py file per exercise and add it to the deposit.
- The figures plotted must also be added to your git repository.
- It is recommended that you spend approximately two weeks on this assignment, as one further assignment sheet will follow.

Exercises:

Exercise 1. Advection equation.

We consider the advection equation in 1D

$$\begin{cases} \partial_t u + \alpha \partial_x u = 0, & x \in [0; 1] \\ u(t, 0) = 0 \\ u(0, x) = u_0(x) \end{cases} \quad (5.1)$$

where α is a constant such that $\alpha > 0$. We set for the present study $\alpha = 1$, $t_{\max} = 0.5$ and

$$u_0(x) = \begin{cases} 100 \exp\left(\frac{0.1}{(x-0.1)(x-0.4)}\right), & 0.1 < x < 0.4 \\ 0, & \text{otherwise.} \end{cases}$$

1. Let u_1 and u_2 two smooth solutions of (5.1). Show that $u_1 = u_2$ by using

$$E(t) = \int_0^1 |u_1(t, x) - u_2(t, x)|^2 dx.$$

2. Deduce that u_{exact} defined by

$$u_{\text{exact}}(t, x) = \begin{cases} u_0(x - \alpha t), & \text{if } x - \alpha t \geq 0 \\ 0, & \text{otherwise,} \end{cases}$$

is the unique smooth solution of (5.1).

3. We now aim at studying numerical schemes simulating PDE (5.1). To do so, we consider the discretization

$$\begin{cases} x_j = j\Delta x, & 0 \leq j \leq N_x & \Delta x = \frac{L}{N_x}, \\ t_n = n\Delta t, & 0 \leq n \leq N_t & \Delta t = \frac{t_{\max}}{N_t} \end{cases}$$

for some $N_x \geq 1$ and $N_t \geq 1$. What are the values of t_0 , t_{N_t} , x_0 , x_{N_x} ?

4. Use the function `Init` from the class `Mesh1D` to plot the discretized initial condition on a mesh containing $N_x = 100$ cells. Implement a function `ExactT(funcExact,t)` in the class `Mesh1D` that return the discretized exact solution at time t . Plot on a graph the initial condition and the exact solution at time $t_{\max}$.

5. We propose the explicit centered scheme:

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} + \alpha \frac{u_{j+1}^n - u_{j-1}^n}{2\Delta x} = 0 \quad (5.2)$$

Compute the theoretical order of the scheme. Perform a von Neumann stability analysis to study the ℓ_2 stability of this scheme. Is this scheme stable?

6. We consider now the explicit upwind scheme:

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} + \alpha \frac{u_j^n - u_{j-1}^n}{\Delta x} = 0 \quad (5.3)$$

- (a) Compute the theoretical order of the scheme. Perform a von Neumann stability analysis to prove that the scheme is ℓ_2 under the CFL condition

$$0 \leq \alpha \frac{\Delta t}{\Delta x} \leq 1. \quad (5.4)$$

- (b) Implement the explicit upwind scheme to compute $u_j^{N_t}$ for all $0 \leq j \leq N_x$ (we will not store the different u^n). Plot on the same graph, the numerical solution u^{N_t} at time $t = t_{\max}$ obtained with $N_x = 20$ and $N_x = 100$ and with $N_x = 200$ and $r = \frac{\alpha \Delta t}{\Delta x} = 0.9$. Add the exact solution. What do you observe?

If this is not done, put the explicit upwind scheme in a function. This function will depend on α , $t_{\max}$, `funcInit`, r and the mesh.

- (c) For $N_x = 100$, we will construct the graph of the ℓ_2 -norm (and ℓ_∞ -norm) of u_h as a function of the time. Is your scheme stable?
- (d) What happens with the numerical solution if the CFL condition (5.4) is not satisfied (for example $r = 1.05$ and $N_x = 50$)? Add the figure obtained on the graph of the previous question. Is your scheme still stable?
- (e) For $N_x = 10$, $N_x = 20$, $N_x = 40$, $N_x = 80$, $N_x = 160$ and $N_x = 320$, we will construct the graph of the error

$$\|u_h^{N_t} - u_{\text{exact}}\|_\infty = \max_{0 \leq j \leq N_x} |u_j^{N_t} - u_{\text{exact}}(t_{N_t}, x_j)|$$

as a function of Δx in log-log scale. What is the numerical order of your finite difference method?

- (f) For $N_x = 10$, $N_x = 20$, $N_x = 40$, $N_x = 80$, $N_x = 160$ and $N_x = 320$, we will construct the graph of the error $\|u_h^{N_t} - u_{\text{exact}}\|_\infty = \max_{0 \leq j \leq N_x} |u_j^{N_t} - u_{\text{exact}}(t_{N_t}, x_j)|$ as a function of Δt in log-log scale. What is the numerical order of your finite difference method?

7. We consider now the implicit upwind scheme:

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} + \alpha \frac{u_j^{n+1} - u_{j-1}^{n+1}}{\Delta x} = 0 \quad (5.5)$$

- (a) Compute the theoretical order of the scheme. Perform a von Neumann stability analysis to study the ℓ_2 stability of the scheme.
- (b) To use this scheme one need to solve a linear system of the form

$$A(u_j^{n+1})_{0 \leq j \leq N_x} = (b_j)_{0 \leq j \leq N_x}$$

Write matrix $A \in \mathcal{M}_{N_x+1}(\mathbb{R})$ and vector $b^n \in \mathbb{R}^{N_x+1}$ using the implicit upwind scheme. Implement this scheme. Plot on the same graph, the numerical solution u^{N_t} at time $t = t_{\max}$ obtained with $N_x = 100$, $r = \frac{\alpha \Delta t}{\Delta x} = 0.9$ and $r = 1.8$. Add the exact solution.

- (c) Perform convergence studies as in Questions 6e and 6f. Add the curves obtained with the implicit upwind scheme on the graph of Questions 6e and 6f.

Exercise 2. Diffusive equation.

We consider the heat equation in 1D

$$\begin{cases} \frac{\partial u}{\partial t} - \alpha \frac{\partial^2 u}{\partial x^2} = 0, & \forall (t, x) \in [0; t_{\max}] \times [0; L] \\ u(t, 0) = 0, & \forall t \in [0; t_{\max}] \quad \text{boundary condition in } x = 0 \\ u(t, L) = 0, & \forall t \in [0; t_{\max}] \quad \text{boundary condition in } x = L \\ u(0, x) = \sin(m\pi x/L), & \forall x \in [0; L] \quad \text{initial condition} \end{cases} \quad (5.6)$$

where u is a function of two variables t and x .

We set for the present study $\alpha = 1$, $m = 3$, $L = 10$, $t_{\max} = 1.2$ and we consider the discretization

$$\begin{cases} x_j = j\Delta x, & 0 \leq j \leq N_x & \Delta x = \frac{L}{N_x}, \\ t_n = n\Delta t, & 0 \leq n \leq N_t & \Delta t = \frac{t_{\max}}{N_t} \end{cases}$$

for some $N_x \geq 1$ and $N_t \geq 1$.

1. What are the values of t_0 , t_{N_t} , x_0 , x_{N_x} ?
2. The explicit Euler (or forward Euler) scheme writes

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} - \alpha \left(\frac{u_{j+1}^n - 2u_j^n + u_{j-1}^n}{(\Delta x)^2} \right) = 0$$

adding discrete analogue of initial and boundary conditions (5.6) leads to the scheme

$$\begin{cases} u_j^{n+1} = u_j^n + \alpha \Delta t \left(\frac{u_{j+1}^n - 2u_j^n + u_{j-1}^n}{(\Delta x)^2} \right), & 1 \leq j \leq N_x - 1, \quad 1 \leq n \leq N_t \\ u_0^n = 0, & 1 \leq n \leq N_t \quad (\text{boundary condition in } x = 0) \\ u_{N_x}^n = 0, & 1 \leq n \leq N_t \quad (\text{boundary condition in } x = L) \\ u_j^0 = \sin(m\pi x_j/L), & 0 \leq j \leq N_x \quad (\text{initial condition in } t = 0) \end{cases} \quad (5.7)$$

Compute the theoretical order of the scheme.

3. Compute the exact solution u_{exact} of (5.6).

4. Prove (on paper) that the numerical scheme is ℓ_∞ stable

$$\forall n, \quad \|u_h^n\|_\infty = \max_{0 \leq j \leq N_x} |u_j^n| \leq \|u_h^0\|_\infty$$

under the CFL condition

$$\frac{\alpha \Delta t}{(\Delta x)^2} \leq \frac{1}{2}. \quad (5.8)$$

5. Perform on the paper a von Neumann stability analysis to get that the scheme is ℓ_2 stable under the CFL condition (5.8).
6. Implement a function `solInit(init,xx)` (where `xx` = $(x_i)_{0 \leq j \leq N_x}$ and `init` is the initial condition function) that returns the initial condition $(u_j^0)_{0 \leq j \leq N_x}$ and a function `solExact(exact,t,xx)` that return the discretized exact solution at time t . Plot on a graph the initial condition and the exact solution at time $t_{\max}$ with $N_x = 100$.
7. Implement the Euler explicit scheme to compute $u_j^{N_t}$ for all $0 \leq j \leq N_x$ (we will not store the different u^n). Plot on the same graph, the numerical solution u^{N_t} at time $t = t_{\max}$ obtained with $N_x = 10$ and $N_x = 20$ and with $N_x = 100$ and $r = \frac{\alpha \Delta t}{(\Delta x)^2} = 0.4$. What do you observe?
If this is not done, put the euler explicit scheme in a function. This function will depend on α , $t_{\max}$, `init`, L and the mesh. and r .
8. What happens with the numerical solution if the CFL condition (5.8) is not satisfied (for example $r = 0.55$ and $N_x = 100$)?
9. For $N_x = 10$, $N_x = 20$, $N_x = 40$, $N_x = 80$, $N_x = 160$ and $N_x = 320$, we will construct the graph of the error

$$\|u_h^{N_t} - u_{\text{exact}}\|_\infty = \max_{0 \leq j \leq N_x} |u_j^{N_t} - u_{\text{exact}}(t_{N_t}, x_j)|$$

as a function of Δx in log-log scale. What is the numerical order of your finite difference method?

10. For $N_x = 10$, $N_x = 20$, $N_x = 40$, $N_x = 80$, $N_x = 160$ and $N_x = 320$, we will construct the graph of the error $\|u_h^{N_t} - u_{\text{exact}}\|_\infty = \max_{0 \leq j \leq N_x} |u_j^{N_t} - u_{\text{exact}}(t_{N_t}, x_j)|$ as a function of Δt in log-log scale. What is the numerical order of your finite difference method?
11. The implicit Euler (or backward Euler) scheme writes for $1 \leq j \leq N_x - 1$, $0 \leq n \leq N_t$:

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} - \alpha \frac{u_{j+1}^{n+1} - 2u_j^{n+1} + u_{j-1}^{n+1}}{(\Delta x)^2} = 0.$$

or equivalently

$$u_j^{n+1} \left(1 + \frac{2\alpha \Delta t}{(\Delta x)^2} \right) - \frac{\alpha \Delta t}{(\Delta x)^2} u_{j+1}^{n+1} - \frac{\alpha \Delta t}{(\Delta x)^2} u_{j-1}^{n+1} = u_j^n \quad 1 \leq j \leq N_x - 1$$

Perform on the paper a von Neumann stability analysis to get that the scheme is unconditionnaly ℓ_2 stable.

12. Initial and boundary conditions are the same as in (5.7). To use this scheme one need to solve a linear system of the form

$$A(u_j^{n+1})_{0 \leq j \leq N_x} = (b_j^n)_{0 \leq j \leq N_x}$$

Write matrix $A \in \mathcal{M}_{N_x+1}(\mathbb{R})$ and vector $b^n \in \mathbb{R}^{N_x+1}$ using the implicit Euler scheme and boundary conditions of (5.7). Implement this scheme.

13. Plot the solution obtained at final time with the implicit Euler scheme with $r = 0.8$ and $N_x = 10$ and $N_x = 100$. Add also the exact solution on the graph.
14. Perform convergence studies as in Questions 9 and 10 with the implicit scheme. What is the numerical order of your finite difference method?
15. The Crank-Nicholson scheme writes for $2 \leq j \leq N_x - 1$, $1 \leq n \leq N_t - 1$:

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} - \frac{\alpha}{2} \frac{u_{j+1}^n - 2u_j^n + u_{j-1}^n}{(\Delta x)^2} - \frac{\alpha}{2} \frac{u_{j+1}^{n+1} - 2u_j^{n+1} + u_{j-1}^{n+1}}{(\Delta x)^2} = 0.$$

Perform on the paper a von Neumann stability analysis to get that the scheme is unconditionnaly ℓ_2 stable.

16. Initial and boundary conditions are the same as in (5.7). To use this scheme one need to solve a linear system. Write this system and implement the Crank-Nicholson scheme.
17. Plot the solution obtained at final time with the Euler implicit and the Crank-Nicholson schemes with $N_x = 10$. Add also the exact solution on the graph.
18. Perform convergence studies as in Questions 9 and 10 with the Crank-Nicholson scheme. What is the numerical order of your finite difference method?